Determining the long-term dynamical stability of exoplanetary systems in the context of habitability using N body simulations

ABSTRACT

Aims: I investigated the long-term habitability of exoplanets in triple-star systems by combining dynamical stability analysis with the evolution of planetary equilibrium temperature and habitable-zone boundaries. I analyzed five observed triple-star systems with known exoplanets: Gliese 667, LTT 1445, HD 132563, Kepler-444, and 94 Ceti. For systems without confirmed terrestrial planets in the habitable zone, I introduced synthetic Earth-like planets to test whether such systems could maintain potentially habitable orbits over long timescales.

Methods: using a Wisdom-Holman style integrator adapted for hierarchical Jacobian coordinates, I explored a broad range of orbital eccentricities and inclinations of the stellar systems and their influence on a 100,000 year timescale. The integrator also tracked planetary temperature variations.

Results: the initial placement within the habitable zone is not sufficient to guarantee long-term habitability. Gliese 667 appears to be the most dynamically stable system with its 2 planets exhibiting relatively moderate temperature variations, making it the strongest candidate under the assumptions of this study. The 94 Ceti and Kepler-444 systems show signs of chaotic dynamical evolution, but some habitable-zone configurations remain thermally stable. LTT 1445 is highly dynamically unstable in many simulated configurations and is therefore unlikely to support long-term habitability under the tested conditions. HD 132563 is dynamically stable in two configurations, but large eccentricity-driven temperature variations make its habitability less likely. These results suggest that combining N-body dynamics with time-dependent temperature and habitable-zone calculations provides a useful framework for evaluating habitability in multi-star exoplanetary systems.

CONCLUSION

This study examined the connection between dynamical stability and long-term habitability in triple-star exoplanetary systems. I implemented a Wisdom-Holman-type symplectic N-body integrator modified for the hierarchical coordinate structure of triple-star systems, and extended it to track planetary equilibrium temperature and habitable-zone boundaries throughout the simulations. The analysis focused on five systems: Gliese 667, LTT 1445, 94 Ceti, HD 132563, and Kepler-444. Gliese 667 contains five known exoplanets, two of which lie within the habitable zone; LTT 1445 contains two confirmed planets and one possible additional habitable-zone candidate; while 94 Ceti, HD 132563, and Kepler-444 do not contain confirmed habitable-zone planets in the adopted dataset.

For systems without known habitable planets, I tested synthetic Earth-like planets placed at the inner edge, midpoint, and outer edge of the calculated habitable zone. Across all five systems, I carried out 101 simulations, each evolved over 100,000 years. The results demonstrate that a planet initially located in the habitable zone is not necessarily dynamically stable, nor does it necessarily remain habitable over long timescales.

The individual systems show substantially different behavior. In 94 Ceti, the synthetic planet placed at the inner edge of the habitable zone was ejected from the system, while other habitable-zone configurations remained more promising. Gliese 667 was the most dynamically stable system in the sample and showed comparatively limited temperature variation, making it the strongest long-term habitability candidate under the assumptions of this work. Kepler-444 exhibited chaotic behavior, including the ejection of one planet, although some synthetic habitable-zone configurations remained stable. LTT 1445 was the least favorable system dynamically: only one configuration remained fully stable, and increasing the eccentricity of the host star's orbit generally made the system more unstable, in some cases leading to the ejection of planets and even the host star. HD 132563 remained dynamically stable for synthetic planets placed near the inner edge and midpoint of the habitable zone, but a planet at the outer edge was ejected. In addition, Kozai-Lidov-type eccentricity oscillations produced temperature variations exceeding 100 K, making long-term surface habitability unlikely without additional climate-stabilizing mechanisms.

Because habitability depends on many factors not included in this model, such as atmospheric composition, internal structure, magnetic shielding, geophysical activity, and climate feedback, these results should not be interpreted as definitive evidence for or against the existence of life. Rather, the simulations identify which systems and orbital configurations are dynamically viable and which maintain relatively stable temperature conditions over long timescales. Future work would incorporate exoplanet climate models, atmospheric effects on equilibrium temperature, and a more detailed treatment of how chaotic orbital evolution affects surface conditions. Extending the same methodology to single-star and binary-star systems would also provide better understanding of long-term habitability across different stellar architectures.